

Appendix A to opinion of ALITO, J.

male or female, or of pertaining to the distinctive function of the male or female in reproduction. *Conjugation*, or *fertilization* (union of germplasm of two individuals), a process evidently of great but not readily explainable importance in the perpetuation of most organisms, seems to be the function of differentiation of sex, which occurs in nearly all organisms at least at some stage in their life history. Sex is manifested in the conjugating cells by the larger size, abundant food material, and immobility of the female gamete (*egg*, *egg cell*, or *ovum*), and the small size and the locomotive power of the male gamete (*spermatozoon* or *spermatozoid*), and in the adult organisms often by many structural, physiological, and (in higher forms) psychological characters, aside from the necessary modification of the reproductive apparatus. Cf. HERMAPHRODITE, 1. In botany the term *sex* is often extended to the distinguishing peculiarities of staminate and pistillate flowers, and hence in dioecious plants to the individuals bearing them.

In many animals and plants the body and germ cells have been shown to contain one or more chromosomes of a special kind (called *sex chromosomes*; *idiochromosomes*; *accessory chromosomes*) in addition to the ordinary paired autosomes. These special chromosomes serve to determine sex. In the simplest case, the male germ cells are of two types, one with and one without a single extra chromosome (*X chromosome*, or *monosome*). The egg cells in this case all possess an *X chromosome*, and on fertilization by the two types of sperm, male and female zygotes result, of respective constitution *X*, and *XX*. In many other animals and plants (probably including man) the male organism produces two types of gametes, one possessing an *X chromosome*, the other a *Y chromosome*, these being visibly different members of a pair of chromosomes present in the diploid state. In this case also, the female organism is *XX*, the eggs *X*, and the zygotes respectively male (*XY*) and female (*XX*). In another type of sex determination, as in certain moths and possibly in the fowl, the female produces two kinds of eggs, the male only one kind of sperm. Each type of egg contains one member of a pair of differentiated chromosomes, called respectively *Z chromosomes* and *W chromosomes*, while all the sperm cells contain a *Z chromosome*. In fertilization, union of a *Z* with a